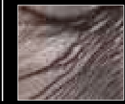
Black holes redefined
Stephen Hawking publishes new paper rectifying certain concepts
http://dgit.in/StepPaper



Frosty dunes on mars
This Wired space photo of the day takes a look at Russell Crater dunes - a favourite for their beauty and ability to depict crucial info about frost accumulation: **http://dgit.in/mdunes**

New Matter, strange Matter

The nuclear age brought with it the realisation that atoms were also divisible. The discovery of sub-atomic particles like ions, electrons, protons and neutrons seemed to settle the matter once and for all. But then there was another revelation, which stands as fact till the present day – all particles are made up of leptons and quarks. In total, all matter seems to be made up of just six leptons, six quarks and 13 other particles that seem to be constantly on the move between leptons and quarks.

These "other" particles are responsible for naturally observed forces such as the electrical forces, which bind all bodies together and the Higgs boson, which gives all particles mass. So by the highly reliable finger counting method, we arrive at a total of 26 particles. And 26 more which are their exact opposites – the antiparticles – in terms of properties such as electric charge.

In attempting to put together a unified and elegant picture of the universe, physicists have created more theoretical particles than real ones they've discovered. But they're determined to close that gap with each passing year. As recently as 2012, experiments at CERN's Large Hadron Collider (LHC) near Geneva successfully observed a particle known as the Higgs boson, just a mere theory in 1964. A more recent finding whose details are still being figured out, was theorised in the Belle experiment conducted at the Japanese High Energy Accelerator Research Organisation, where scientists searching for the anomalous (meaning, it doesn't fit into their current picture of the universe) Y(4260) particle ended up discovering the theoretical particle Y(3600).

Even with the discovery of all these new particles, there remains a category of matter, which only the bravest scientists take on: exotic matter. Any matter that doesn't behave according to normal predictable patterns is termed as "exotic matter". And in terms of sheer quantity, there's a lot of it.

What is exotic matter?

Physicists have spent decades trying to arrive at a deterministic understanding of how things function. Determinism basically means that if all the rules are known and all the particles are observed, we can

Cosmic bodies are just like balls on a pool table, observing the same rules - as long as there are no black holes

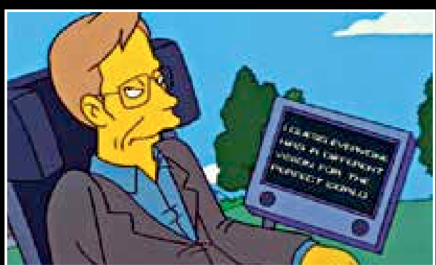
predict how everything is going to interact – we'll be omniscient about cause and effect.

Imagine a pool table, with the different coloured balls as different particles; if you know all the facts of the table – mass, velocity, gravitational constant, etc – you can predict how each strike of the ball will alter the positions of the balls on the table using simple Newtonian physics. This same principle allows us to measure astronomical phenomenon, its determinism. And physicists are losing sleep over it.

Strange ideas about matter

According to theoretical physics, all of reality can be reduced to mathematical models. If the mathematical equations are sound, then reality must conform to its predictions. It's not faith, not belief, it's cold, hard logic. And theoretical physicists make a living formulating these equations of certainty. These are the same equations that earn prizes and awards and help us achieve accomplishments such as space travel or manufacturing of cutting edge electronics.

Without them, we'd be lost, or at least a lot more uncertain of the consequences of our scientific actions. These models take into account all observable phenomenon and present them as related elements of energy causality; in simple terms, they tell us the way all energy and matter interact.



UNIFIED THEORY OF EVERYTHING – If it was easy, you'd be doing it

And since all energy is matter and vice versa, these equations have mathematical "remainders" or "constants" that need to exist for the equations to be valid.

In this manner, physicists allowed for different forms of matter, whether observed in nature or not, to be a part of reality. As long as the equations demand their existence, it is accepted that they must exist. However, as scientists have continued to broaden their perspectives from beyond the scale of the Earth into the distant reaches of space they've come across some troublesome issues causing them to cast doubt on all their models. The issues pertain to standarising the rules of the planets and stars to make them correspond with the rules of the electrons and atoms – unifying the models of General Relativity and Quantum Mechanics.

General Relativity vs. Quantum Mechanics

We all know how to divide fractions using basic mathematics – numerator divided by denominator results in a quotient. And sometimes a remainder. Most of the times it's a simple fraction to reasonable decimal points; other times it reaches the infinity of pi. But it exists as a specific function on the number line. But when we deal with theoretical equations of matter and energy, the remainders aren't nearly as simple.

These remainders are what we term as exotic matter – those parts of reality without which the equation loses all validity. The two models of quantifying reality are split between General Relativity and Quantum Mechanics. Bridging these two models and creating one unified theory of everything (TOE) is the holy grail of Physics.

General Relativity is the descendent of Newtonian Physics (The concepts and equations that describe the precise relationship between all bodies, especially those we see in space). Albert Einstein's lucid theories of general relativity lay bare the nature of the universe and explain how cosmic bodies relate to each other within the fabric of space-time and gravity. Quantum Mechanics, on the other hand, operates on fundamentals that reject all of general relativity's predictions. It deals with the tiniest parts of the universe, that is the atomic structure, subparticles and